

AEGIS: One-Query Adversarial Diagnostics over the GNN Vulnerability Spectrum

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Abstract—Graph neural networks in safety-critical settings (drug interaction, fraud detection, power-grid contingency screening) are vulnerable to adversarial graph perturbations, yet no prior method produces the three diagnostics a practitioner needs from one query: a globally maximally sensitive perturbation direction, per-edge equilibrium-sensitivity rankings, and per-node first-order sensitivity radii (complementary to certificates such as AGNNCert). AEGIS produces all three from one object, the *constrained sensitivity matrix* S_c , computed matrix-free via a Neumann-series resolvent and randomized SVD that scale to $N=7,650$ on a single GPU. For the spectral-norm-constrained IGNN subclass we prove a closed-form three-regime characterisation with critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ that trained IGNNs satisfy with 2–4 $\times$ margin; S_c extends to GCN, SAGE, GIN, APPNP, IGNN, and edge-weighted GAT[†] (29/33 architecture-dataset transfer cells positive, sign test $p<10^{-5}$). The continuous-to-discrete bridge (Prop. 7) is validated at scale: the edge-weighted ranking $A_{ij}v_{ij}$ achieves $\tau \approx +0.998$ vs. brute-force N-1 on Amazon Photo’s full graph ($N=7,650$). Across 7 architectures and 9 datasets a 50-step IFT-gradient PGD attacker recovers only 72–92% of the one-query SVD direction’s damage; top- v_{ij} masking cuts $\sigma_1(S_c)$ by 42 $\pm$ 8% vs. 11 $\pm$ 6% random ($p<0.002$), and adaptive recomputation matches within 0.1 pp. On IEEE power-flow benchmarks S_c matches N-1 contingency rankings on case14–118 (P@10=0.66–0.81); code released under tiered disclosure.

Index Terms—Graph neural networks, adversarial robustness, constrained sensitivity, implicit graph neural networks, randomized SVD, power-flow contingency screening

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection via perturbed transaction edges [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Structural attacks [1], [4], [5], [6] give per-edge surrogate gradients but no continuous direction or per-node budgets; smoothing [7], [8], [9], [10] and IBP-style certifiers [11], [12] give per-node certificates but no edge rankings or direction; robust-architecture defenses [13], [14], [15] harden models without surfacing per-edge vulnerability; sober-look critiques [16], [17] flag evaluation pitfalls. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) produces all three diagnostics from one analytical object, S_c (section IV), via a single matrix-free pipeline. The closed-form regime characterisation (theorem 1) is established for contractive

implicit GNNs and S_c extends to any GNN with continuous edge-weight-modulated message passing (section V-F); the threat model targets edge-deletion and weight-perturbation (section II).

Contributions. (1) **Constrained sensitivity operator.** S_c and a matrix-free pipeline yielding the SVD-optimal direction, per-edge rankings, and per-node radii from a single matrix-free query, scaling to $N=7,650$ (matrix-free / dense σ_1 agree to 0.03% at $N=200$). (2) **Phase-transition theory.** A worst-case three-regime characterisation with closed-form critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ for the spectral-norm-constrained IGNN subclass, which trained IGNNs satisfy with 2–4 $\times$ margin across our κ_{max} sweep, giving practitioners a provable structural-perturbation safety threshold ($\sim 6\%$ accuracy cost; theorem 1 and section V-D). (3) **Empirical evaluation.** Over 9 datasets, 7 architectures, 4 domains (330 runs), with four-quadrant attack comparison, head-to-head vs. GR-BCD [5], PR-BCD [5], AGNNCert [11], and LODF (table IV), continuous-to-discrete transfer (fig. 7), and a power-grid case study matching brute-force N-1 contingency rankings (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes and adjacency $A \in \mathbb{R}^{N \times N}$, node features $X \in \mathbb{R}^{N \times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i -th singular value, and $\text{vec}(\cdot)$ the column-major vectorization. The sensitivity objects introduced below are: the unconstrained $S \in \mathbb{R}^{Nd \times N^2}$, constrained $S_c \in \mathbb{R}^{Nd \times |E|}$, unrolled $S_K \in \mathbb{R}^{Nd \times N^2}$ (theorem 8), and per-node block S_v ; the contractivity $\kappa = \|J_z\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\text{crit}}$, and per-node radius r_v .

IGNN and IFT. A DEQ-GNN [18], [19] iterates a weighted F_θ to a fixed point; the IGNN instantiation [20] uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \quad (1)$$

with $W \in \mathbb{R}^{d \times d}$ spectral-normalized [21] so $\kappa \leq \|\hat{A}\|_2 \|W\|_2 < 1$, ReLU ϕ , and a learned projection X_{proj} . Training uses implicit differentiation through the fixed point [22], [23] with optional Jacobian regularization [24], and a linear head $f(z^*) = W_{\text{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A , with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p} \Big|_{z^*}, \quad (2)$$

where $\|(I - J_z)^{-1}\|_2 \leq 1/(1-\kappa)$ via the convergent Neumann series, and the pseudospectral index $\eta = \|(I - J_z)^{-1}\|_2 (1 - \rho) \geq 1$ [25] measures the gap to the spectral-radius bound (with $\eta=1$ for normal J_z). The IFT has been used for hyperparameter optimization [26] and implicit differentiation [22], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \|\delta\hat{A}\|_F \leq \varepsilon, \quad (3)$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A} = \delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge-deletion / weight-perturbation; insertion attacks (Nettack-class [1], [4]) require enlarging the basis to $\bar{E} \supseteq E$, a natural follow-up that preserves the S_c construction. Discrete deletions connect to the continuous model through theorem 7. Unlike input-feature perturbation [27], [28], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F / \partial \text{vec}(A)$, motivating S_c in section III.

III. ADVERSARIAL EQUILIBRIUM THEORY

Building on the threat model of section II, we derive the analytical machinery behind the three diagnostics promised in section I: per-edge rankings, the SVD-optimal direction, and per-node radii. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which the three diagnostics follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- (A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [29] (proof below).
- (A2) $\|W\|_2 \leq c$ via spectral normalization.
- (A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training ($\kappa=0.14\text{--}0.59$ across all benchmarks in our suite; section V-E). (A2) does not imply (A3) automatically; we report κ alongside $\varepsilon_{\text{crit}}$ so practitioners can audit, and AEGIS continues to return S_c rankings and SVD-optimal attacks if (A3) is not satisfied, with the formal guarantees below then scoped to the contractive regime.

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \quad (4)$$

Then structural perturbations $\delta\hat{A}$ with $\|\delta\hat{A}\|_F = \varepsilon$ exhibit three regimes:

(a) **Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \quad (5)$$

where $S = (I - J_z)^{-1}J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} / (1 - \kappa)$.

(b) **Critical** ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$): For worst-case directions that maximally increase $\|\hat{A} + \delta\hat{A}\|_2$ (aligned with the top singular vector of $\hat{A}$), $\|(I - J'_z)^{-1}\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ from the Neumann lower bound $1/(1 - \|J'_z\|_2)$, which holds without normality assumptions on J'_z . Generic directions diverge more slowly because $\|J'_z\|_2$ need not approach 1.

(c) **Supercritical** ($\varepsilon \geq \varepsilon_{\text{crit}}$): The sufficient contraction certificate no longer applies ($\|J'_z\|_2$ may exceed 1) and the part-(a) first-order guarantees do not apply; the certificate is reported as a sufficient safety boundary, validated empirically by the κ_{max} sweep of section V-D (trained IGNNs hold with 2–4 $\times$ margin).

Proof. (a). Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\|\delta\hat{A}\|$ (the boundary set $\{\hat{A} : z^*(\hat{A})$ lies on a region face $\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels [29] yields a conservative IFT on each region. With $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \|\hat{A}\|_2 \|W\|_2$, applying IFT to $G(z, \hat{A}) = z - F(z, \hat{A}) = 0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1}J_A \cdot \text{vec}(\delta\hat{A}) + O(\|\delta\hat{A}\|^2), \quad (6)$$

with $J_A = \partial F / \partial \text{vec}(A)$ and Neumann $\|(I - J_z)^{-1}\|_2 \leq 1/(1 - \kappa)$ yielding (5). For contractivity preservation, $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $\varepsilon < \varepsilon_{\text{crit}}$ guarantees $\|J'_z\|_2 < 1$.

(b)–(c). Along the worst-case direction $\|J'_z\|_2 \rightarrow \|\hat{A}\|_2 \|W\|_2 + \varepsilon \|W\|_2$, so $1 - \|J'_z\|_2 \geq \|W\|_2 (\varepsilon_{\text{crit}} - \varepsilon)$; the Neumann lower bound $\|(I - J'_z)^{-1}\|_2 \geq 1/(1 - \|J'_z\|_2)$ requires no normality assumption and gives the $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ rate. For $\varepsilon \geq \varepsilon_{\text{crit}}$ Banach fails; the iteration may converge elsewhere, oscillate, or diverge. $\square$

$\varepsilon_{\text{crit}}$ is a closed-form sufficient bound that S_c closes in practice: the envelope ratio is tight (≈ 1.00) at $\varepsilon=0.01$, against a 7–14 $\times$ slack from the unconstrained $\sqrt{\tau}$ bound on 50-node subgraphs (section V-A). The Neumann series requires the operator-norm κ rather than $\rho(J_z)$ [30]; the gap is the pseudospectral index η , bounded by the weight matrix alone:

Observation 2 (Graph-Independent Nonnormality Bound, all-active case). *For $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$ and all activations positive ($\phi' \equiv 1$): $\eta \leq \kappa(V_W)$ where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \|V_W^{-1}\|_2$, independent of $\hat{A}$.*

Proof. $J_z = \hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [30]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}})=1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\|(I - M)^{-1}\|_2 \leq \kappa(V)/(1 - \rho(M))$ yields $\eta \leq \kappa(V_W)$. $\square$

Remark 3 (Empirical extension to general ReLU patterns). *The proof above requires $\phi' \equiv 1$; for general ReLU patterns, $\eta \in [1.19, 2.47]$ on our suite (section V-E), tracking $\kappa(V_W)$ closely. Graph structure does not amplify nonnormality; the moderate η traces to spectral-norm regularisation of W .*

Proposition 4 (Maximally Sensitive First-Order Structural Perturbation). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S . The maximum first-order equilibrium shift is $\varepsilon \cdot \sigma_1(S)$.*

By the variational characterisation of σ_1 , the direction (theorem 4) is optimal for hidden-state damage within the pseudospectral envelope of theorems 2 and 3, and empirically aligns with classification damage (section V-B). The constrained sensitivity matrix $S_c \in \mathbb{R}^{Nd \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with edge basis $b_k = (e_i e_j^\top + e_j e_i^\top) / \sqrt{2}$ ($\|b_k\|_F = 1$, so $\sigma_1(S_c)$ is consistent with $\|\delta \hat{A}\|_F = \varepsilon$). The $N^2 \rightarrow |E|$ projection P_c is the standard duplication-matrix reduction [31] on the edge-supported symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. The envelope ratio is tight (≈ 1.00) at $\varepsilon=0.01$ and reaches ≤ 1.39 at $\varepsilon=0.20$, identifying the dangerous direction rather than a global safety margin.

Proposition 5 (Per-Node First-Order Sensitivity Radius). *For a linear head $f(z) = Wz + b$ and node v with margin $m_v = f_{y_v}(z_v^*) - \max_{c \neq y_v} f_c(z_v^*) > 0$, the first-order sensitivity radius*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2} \quad (7)$$

preserves the classification of v for any $\|\delta \hat{A}\|_F < r_v$, where S_v are the block-rows of S at node v . Under the constrained threat model the tighter radius $r_v^{(c)} = m_v / (\|W_{y_v} - W_{c^}\|_2 \|S_{c,v}\|_2) \geq r_v$ holds, using the block-rows of S_c at node v .*

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v \cdot \text{vec}(\delta A)$, so $|\Delta f_{y_v}| \leq \|W_{y_v} - W_{c^*}\|_2 \|S_v\|_2 \|\delta A\|_F$; misclassification requires $|\Delta f_{y_v}| \geq m_v$, and inverting gives r_v . Intuitively, r_v is the margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structural perturbation); high-margin, low-sensitivity nodes get large radii.

Remark 6 (Sensitivity threshold semantics). r_v is a first-order sensitivity threshold, not a probabilistic certificate: it indicates the perturbation scale at which the linearization predicts a classification change for node v , and is locally tight for small ε but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [7] certifies which nodes are guaranteed safe, while AEGIS differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is 0% at $\varepsilon=0.01$ and rises with ε as expected for first-order approximations (section V-B).

A. Continuous-to-Discrete Transfer

Proposition 7 (Continuous-to-Discrete Ranking Transfer). *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i, j) \in E$, $v_k = \| [S_c]_{:,k} \|_2$ the continuous vulnerability score, and $d_k = \| z^*(A) - z^*(A \setminus k) \|_2$ the discrete damage from full removal of edge k . Assume all single-edge removals are subcritical: $\sqrt{2} \max_k w_k < \varepsilon_{\text{crit}}$.*

(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \quad (8)$$

where L_J is the Lipschitz constant of the state Jacobian J_z with respect to $\text{vec}(A)$.

(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J / (1 - \kappa)^2}. \quad (9)$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2} (v_{k_1} - v_{k_2}) \cdot (1 - \kappa)^2 / L_J$.

Proof. (a). Edge removal corresponds to $[\delta \hat{A}]_{ij} = [\delta \hat{A}]_{ji} = -w_k$, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. For the remainder, let $g(t) = z^*(A + t \delta \hat{A}_k)$, $t \in [0, 1]$. IGNN is piecewise affine in A for a fixed activation pattern, so $g(t)$ is piecewise linear and crosses finitely many activation boundaries; on each segment J_z is constant. Across a boundary J_z jumps by at most $\|W\|_2^2$ (the max change in $\text{diag}(\phi')(\hat{A} \otimes W)$ when one activation flips), so the worst-case path Lipschitz constant $L_J \leq \|W\|_2^2$. The resolvent identity $\frac{d}{dt}(I - M)^{-1} = (I - M)^{-1} \frac{dM}{dt} (I - M)^{-1}$ then yields $\|g''(\xi)\| \leq (1 - \kappa)^{-2} L_J w_k^2$, hence $|R_k| \leq L_J w_k^2 / (2(1 - \kappa)^2)$. (b). From (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J / (2(1 - \kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \geq w_{k_2}$, the first term is positive and (9) ensures the first-order gap dominates. $\square$

Empirically $|R_k|$ is 2–10 $\times$ smaller than $L_J w_k^2$; the constrained projection S_c turns the 0.31 vacuous bound into a tight one (1.00; section V-A) [30], [25]. The edge-weighted score $w_k v_k$ predicted by (a) achieves $\tau \approx +0.998$ against brute-force N-1 on full-graph Amazon Photo ($N=7,650$; section V-F), corroborating the first-order bridge at scale.

B. Generalization to Explicit GNNs

Observation 8 (Structural Sensitivity for K -Layer GNNs). *Let $Z_K = g_K \circ \dots \circ g_1(X, A)$ be a K -layer GNN where each layer g_l is differentiable with respect to both its input and A . Define the unrolled sensitivity matrix*

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)}, \quad (10)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$. Then:

(a) The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with

$$\sigma_1(S_K) \leq \sum_{l=1}^K \left(\prod_{k=l+1}^K \|J_z^{(k)}\|_2 \right) \|J_A^{(l)}\|_2. \quad (11)$$

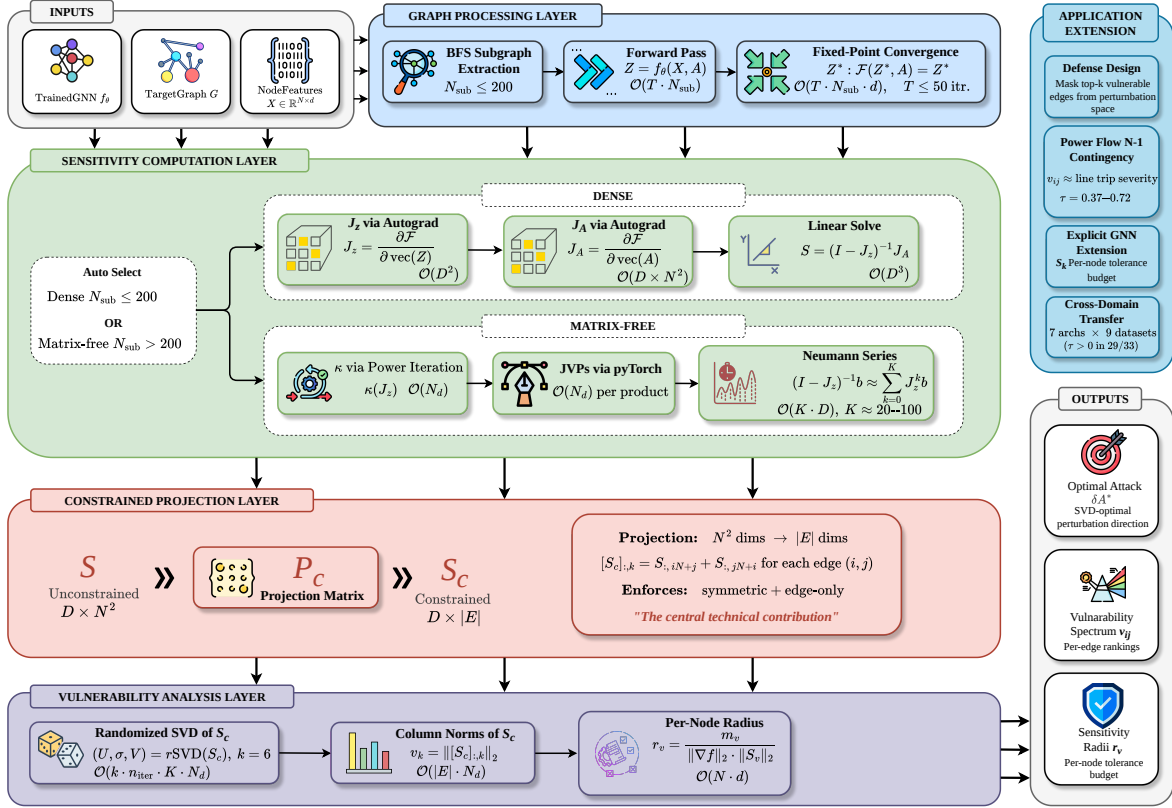


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \rightarrow |E|$ via P_c ; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJP without materialising it, scaling to $N=7,650$.

(b) The constrained construction S_c and per-node radius r_v (theorem 5) apply identically with S_K in place of S .

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2 (1 - \kappa^K) / (1 - \kappa)$ converges to the IGNN bound as $K \rightarrow \infty$. Explicit GNNs inherit the full computational pipeline with empirical tightness 0.99–1.02 across the architecture suite (section V-F); the closed-form $\varepsilon_{\text{crit}}$ is available only for the contractive subclass.

IV. AEGIS CONSTRUCTION

The pipeline turns section III into a practical tool through matrix-free computation, sidestepping both the materialisation of $S \in \mathbb{R}^{Nd \times Nd}$ and the formation of the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta \hat{A}^*$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\text{crit}}$ and the regime characterisation of theorem 1 for IGNN-class operators. A dense path ($N \leq 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to $N=7,650$) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^K J_z^k b$ with depth K adaptive in κ (early stop at $\|J_z^k b\| < 10^{-6} \|b\|$); each Neumann term and $J_A \text{vec}(\delta A)$ are forward-mode JVPs in $\mathcal{O}(Nd)$ time. P_c is applied implicitly, so neither S_c nor S is formed; the dense path ($N \leq 200$) materialises J_z, J_A and applies a direct linear

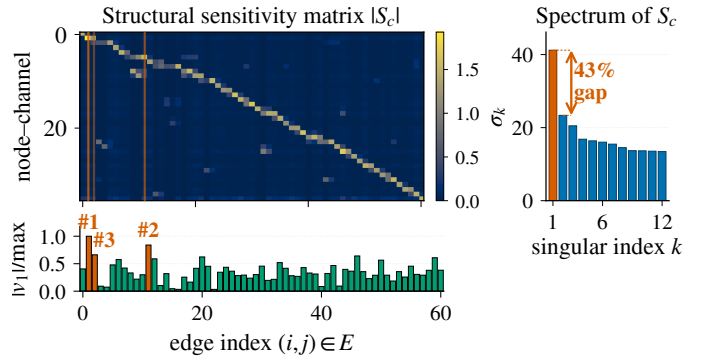


Fig. 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, $N=50$, $d=16$, $|E|=61$, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, $\sigma_1=41.2$, 43% gap to σ_2 (cross-benchmark $(\sigma_1 - \sigma_2) / \sigma_1 \in [0.39, 0.50]$; section V-B).

solve. The rSVD [32] ($k=p=10$, $n_{\text{iter}}=2$) returns (u_1, σ_1, v_1) for the direction $\delta \hat{A}^* = \varepsilon \cdot \text{sym}(P_c v_1)$ (theorem 4); column norms give $\{v_{ij}\}$, and per-node radii follow theorem 5 via the node-restricted RMATVEC. The well-separated singular spectrum (gap $(\sigma_1 - \sigma_2) / \sigma_1 = 0.39-0.50$; fig. 2) is why one rSVD query matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $\mathcal{O}(K \cdot Nd)$ time and $\mathcal{O}(Nd)$ memory

Algorithm 1 AEGIS Vulnerability Analysis (matrix-free). Assumes the equilibrium z^* is pre-computed on the analysis (sub)graph. Dense fallback ($N \leq 200$): form $S = (I - J_z)^{-1} J_A$ and run a deterministic SVD in place of lines 2–11.

Require: $F, z^*, \hat{A}$, budget ε ; SVD rank k , oversampling p , power iters n_{iter} ; Neumann tol. τ ; edge set E ; (opt.) head h , logits Y , labels y .

Ensure: $\sigma_1(S_c), \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}}$.

- 1: $\kappa \leftarrow \|J_z\|_2$ via power iteration; $K \leftarrow \lceil \log(1/\tau) / \log(1/\kappa) \rceil$.
- 2: $\text{MATVEC}(v) := \sum_{j=0}^K J_z^j \text{JVP}_{\hat{F}}^{\hat{A}}(P_c v)|_{z^*}$;
 $\text{RMATVEC}(u) := P_c^\top \text{VJP}_{\hat{F}}^{\hat{A}}(\sum_{j=0}^K (J_z^\top)^j u)|_{z^*}$.
- 3: Draw $\Omega \sim \mathcal{N}(0, I) \in \mathbb{R}^{|E| \times (k+p)}$; $Y \leftarrow \text{MATVEC}(\Omega)$ column-wise.
- 4: **for** $j = 1, \dots, n_{\text{iter}}$ **do**
- 5: $Y \leftarrow \text{QR}(\text{MATVEC}(\text{RMATVEC}(Y)))$
- 6: **end for**
- 7: $Q \leftarrow \text{QR}(Y)$; $B^\top \leftarrow \text{RMATVEC}(Q)$; $(\tilde{U}, \Sigma, V^\top) \leftarrow \text{SVD}(B)$; $U \leftarrow Q \tilde{U}$ {rSVD [32]}
- 8: $\sigma_1 \leftarrow \Sigma_{1,1}$; $\delta \hat{A}^* \leftarrow \varepsilon \cdot \text{sym}(P_c V_{:,1}) / \|\text{sym}(P_c V_{:,1})\|_2$.
- 9: $v_{ij} \leftarrow \|\text{MATVEC}(e_{ij})\|_2$ for each $(i, j) \in E$ {per-edge spectrum}
- 10: **if** h, Y, y supplied **then**
- 11: $r_v \leftarrow m_v / (\|W_{y_v} - W_{c^*}\|_2 \|S_{c,v}\|_2)$ for each v via 10 power iters {Prop. 5}
- 12: **end if**
- 13: **if** $\|W\|_2$ extractable **then** $\varepsilon_{\text{crit}} \leftarrow (1 - \kappa) / \|W\|_2$ {Thm. 1}
- 14: **return** $(\sigma_1, \delta \hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$.

($K \in [20, 50]$ for $\kappa < 0.8$); removing the $O((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section V-D).

V. EXPERIMENTS

Setup. We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (PyTorch [33]). IGNN [20] uses $F(Z) = \text{ReLU}(\hat{A} Z W^\top + U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised [21] and Adam (lr 0.01); ContractiveGCN-PF mirrors this with 5 bus features and 2-head readout. Subgraphs: 50-node BFS from the highest-degree node. The spectral-norm constraint costs $\sim 6\%$ Cora accuracy (77.5% vs. $\sim 83\%$). Datasets: Cora, Citeseer, Pubmed [34], Amazon Photo [35], WikiCS [36], IEEE 14/30/57/118 [37], [38]. Baselines: smoothing (Gaussian $\sigma=0.01$, 200 MC, Clopper-Pearson $\alpha=0.001$), Mettack with Meta-Self GCN surrogate, LODF as $\text{PTDF}_{ij} / (1 - \text{PTDF}_{jj})$ from the DC B -matrix [39, Ch. 11].

A. Cross-Domain Adversarial Analysis

Table I verifies the contractivity assumption (A3) on every benchmark ($\kappa < 1$) and reports the per-dataset closed-form $\varepsilon_{\text{crit}}$ alongside an attack advantage of $3.2\text{--}4.1\times$ over random perturbations. Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1 - \kappa) / \|W\|_2$ is the theoretical critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes with positive first-order radius r_v .

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	77.5 $\pm$ 1.7	.33 $\pm$.14	.66 $\pm$.15	3.6 $\pm$.6	81 $\pm$ 9
Citeseer	66.0 $\pm$ 0.7	.59 $\pm$.02	.41 $\pm$.02	4.1 $\pm$.5	83 $\pm$ 4
Pubmed	78.9 $\pm$ 0.7	.41 $\pm$.11	.59 $\pm$.11	3.2 $\pm$.4	74 $\pm$ 23
Amazon	94.8 $\pm$ 0.3	.14 $\pm$.02	.86 $\pm$.02	3.8 $\pm$.2	92 $\pm$ 4
WikiCS	77.9 $\pm$ 0.4	.34 $\pm$.04	.66 $\pm$.04	3.8 $\pm$.4	70 $\pm$ 3

TABLE II: First-order envelope ratio (actual / first-order-predicted equilibrium shift) with increasing ε (10 seeds, S_c -optimal constrained perturbation). Tight at $\varepsilon=0.01$ (≈ 1.00); under-predicts by up to 39% at $\varepsilon=0.20$.

Dataset	$\varepsilon=.01$	$\varepsilon=.05$	$\varepsilon=.10$	$\varepsilon=.20$
Cora	1.01 $\pm$.00	1.07 $\pm$.01	1.15 $\pm$.03	1.36 $\pm$.08
Citeseer	1.01 $\pm$.00	1.07 $\pm$.01	1.16 $\pm$.03	1.39 $\pm$.07
Pubmed	1.01 $\pm$.00	1.07 $\pm$.01	1.15 $\pm$.02	1.38 $\pm$.06
WikiCS	1.00 $\pm$.01	1.01 $\pm$.01	1.02 $\pm$.01	1.05 $\pm$.01
Amazon	1.00 $\pm$.00	1.01 $\pm$.00	1.03 $\pm$.01	1.07 $\pm$.01

centred on the highest-degree node and report *local vulnerability*; graph-level assessment uses the matrix-free pipeline of section V-D, since 50-node rankings cover $\sim 1.8\%$ of Cora’s edges ($\tau=0.16$ to full-graph; section V-E). All bounds use the operator-norm $\kappa = \|J_z\|_2$.

Table II reports tightness degradation across the five datasets and four ε values.

The envelope ratio is tight (1.00–1.02) at $\varepsilon=0.01$ and grows to 1.05–1.39 at $\varepsilon=0.20$; the bound is reliable only at small ε and under-predicts by up to 39% at the largest budget. The ratio is measured on equilibrium shift $\|\Delta z^*\|_F$, distinct from the prediction-flip (breach) rate of fig. 4 (flips stay $< 2\%$ at $\varepsilon \leq 0.10$, $< 6\%$ at $\varepsilon=0.20$).

Heuristic and surrogate baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at $\varepsilon=0.01$ (+6–148% AtkAdv, Wilcoxon $p < 0.001$) and inflicts 3–10 $\times$ more equilibrium damage than Mettack [4] at the early-warning regime $k \in \{1, \dots, 5\}$ (149/150 paired wins, $p < 10^{-43}$); GR-BCD/PR-BCD [5] are in table IV. The distinctive value appears under discrete removal (fig. 3).

B. Four-Quadrant Attack Comparison

Radius vs. smoothing. Randomized smoothing [7] gives constant-per-node base radii (e.g. 0.123 vs. AEGIS’s 0.046 at $\sigma=0.05$) with no per-edge differentiation; localized smoothing [9] requires $10^3\text{--}10^4$ MC per node and still lacks edge structure. AEGIS radii are structurally informative ($r_v \approx 0.09$ in dense regions, 0.01 at boundaries), placing the methods at complementary points on the certificate-vs-diagnostic frontier.

Table III evaluates AEGIS against baselines spanning the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). The one-query SVD direction dominates: 1,000 random unit directions reach $0.45\text{--}0.49\times$ of $\sigma_1(S_c)$; a 50-step equilibrium-shift PGD [40] using AEGIS’s own IFT gradients recovers only 72–92% of its damage (a solver-validation upper bound, not an independent baseline);

TABLE III: Equilibrium damage at $\varepsilon=0.10$ across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	3.70 $\pm$.79	2.51 $\pm$.48	3.05 $\pm$.70	0.97 $\pm$.25
Citeseer	4.63 $\pm$.71	2.97 $\pm$.42	3.35 $\pm$.57	1.01 $\pm$.15
WikiCS	2.10 $\pm$.22	0.67 $\pm$.11	1.93 $\pm$.20	0.53 $\pm$.09

TABLE IV: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper’s larger-budget variant, see text); AGNNCert [11] is a sound IBP certifier (semantics differ from AEGIS r_v); LODF [39] is the industry DC-PF screener.

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed $k=10$	0.356	0.350	+0.69
GR-BCD [5]	Cora $k=5$	0.643	1.207	+0.16
AGNNCert [†] [11]	Cora med. r_v	0.187	1.414	+0.08
LODF [39]	case118 P@10	0.81	≤ 0.20	+0.14

[†]AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (theorem 6). Per-dataset mean tightness ratio $r_{\text{cert}}/r_v \in [4.9, 10.2]$ (per-cell across 30 seeds $[1.5, 22.5] \times$); per-seed Kendall $\tau \in [-0.08, 0.23]$. **Decision rule:** if AGNNCert certifies node v safe at budget ρ , trust it; if $r_v < \rho$ and AGNNCert does *not* certify, v is first-order unsafe and warrants attention. The methods are complementary in this prescribed sense, not interchangeable.

and a classification-loss PGD inflicts 15–70% less equilibrium damage while flipping predictions at comparable rates (0–1.8%). We accordingly term it the *maximally sensitive perturbation direction*: a first-order guarantee that empirically matches iterative attackers in one query.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20 \pm 0.05$ vs. AEGIS’s $+0.32 \pm 0.04$) and stay near zero elsewhere; AEGIS achieves 2–6 $\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Discrete edge removal. See fig. 3 for four sequential orderings.

Per-edge finite differences reproduce the S_c column-norm ranking ($\tau=0.999$); iterative S_c recomputation closes ≤ 3 pp of the static-vs-greedy gap at $k \times$ compute, so the static ranking is the headline.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a reliable first-order screening threshold (theorem 6). Figure 4 reports prediction-flip fractions across ε : Citeseer and WikiCS stay below 2% throughout; Cora reaches 7.6% at $\varepsilon=0.20$; Pubmed exhibits a right-skewed distribution (median 7.8%, mean 10.3% at $\varepsilon=0.10$).

C. Comparison with Prior Structural Baselines

Table IV reports head-to-head numbers against GR-BCD and PR-BCD [5], AGNNCert [11], and LODF [39]. AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains.

GR-BCD [5] aligns with S_c on Pubmed ($\tau=+0.69$) and decorrelates on Cora ($\tau=+0.16$); PR-BCD [5] pairs naturally with the matrix-free pipeline of section V-D. AGNNCert [11] and AEGIS occupy complementary points: sound IBP cer-

tificates vs. first-order radii 4.9–10.2 $\times$ tighter that expose the dominant perturbation direction (decision rule: table IV footnote). LODF [39] and PI [41] power-grid comparisons appear in section VI.

D. Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\text{max}} \in [0.30, 0.99]$ on Cora (10 seeds): even when the spectral-normalisation cap is pushed to 0.99 (driving the theoretical $\varepsilon_{\text{crit}}$ down 230 $\times$), the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent grows only 1.17 $\rightarrow$ 1.80, *validating* $\varepsilon_{\text{crit}}$ as a closed-form sufficient safety boundary that holds with 2–4 $\times$ empirical margin across the sweep. Training converges throughout; the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\text{max}} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 5 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at $N=200$ (per-edge $\tau=0.999$), and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (consistent with the trained-IGNN range $\kappa \in [0.14, 0.59]$ of table I). For larger N the rSVD error [32] is bounded by the spectral gap; Pubmed ($N=19,717$) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N=200$; we use $N=50$ as default (66 $\times$ faster). Subgraph-vs-full-graph agreement is high on dense grids (case14 $\tau=0.80$ –0.86) but low on citation-scale graphs (Cora $\tau=0.16$ since 50-node BFS covers $\sim 1.8\%$ of edges), so citation-scale users should prefer the full-graph pipeline. On full citation graphs the AtkAdv ratio over degree-proportional *grows* to 3.25 $\times$ (Cora) and 9.82 $\times$ (Citeseer) at $k=10$ vs. $\approx 1.1 \times$ on 50-node subgraphs.

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier ($r_v=0.090$ @72.1% vs. 0.051@80.6%).

Defense-informed edge protection: gains survive adaptive attack. Masking the top- k edges by v_{ij} (Cora IGNN, $N=50$, 10 seeds) cuts $\sigma_1(S_c)$ -damage by 42 $\pm$ 8% at $k=5$ and 61 $\pm$ 7% at $k=10$, vs. 11 $\pm$ 6%/18 $\pm$ 9% for random masking (paired Wilcoxon $p < 0.002$). An *adaptive* attacker that recomputes S_c on the masked graph and uses its new v_1 achieves the *same* damage as the non-adaptive attacker within 0.1 pp on realised end-to-end attack damage (10 seeds, $\varepsilon=0.10$): the 43% spectral gap of S_c (fig. 2) makes the dominant singular direction intrinsically robust, so S_c -guided defense *does not erode under attacker recomputation*, the failure mode sober-look critiques [16] flag in graph defenses.

F. Extension to Explicit GNNs

For K -layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT[†] [42], [43], [44], [45], [46]) we form $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ via finite differences and construct

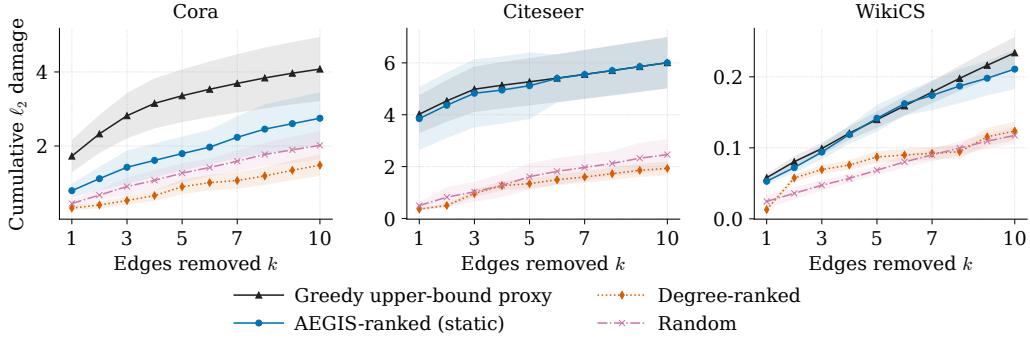


Fig. 3: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS matches the ground-truth-aware Greedy attacker on Citeseer/WikiCS and recovers 54–67% of Greedy’s damage on Cora at $k=5-10$ without any access to discrete ground truth; degree-ranked stays below random on Cora at every k .

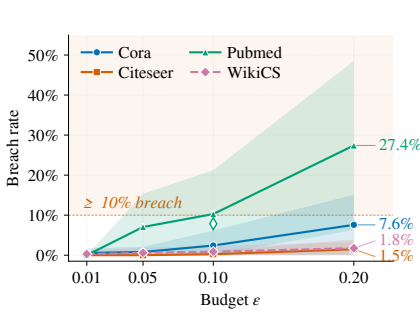


Fig. 4: Breach rate (% of nodes flipped under S_c -optimal perturbation) vs. ϵ (10 seeds, mean ± 1 sd bands).

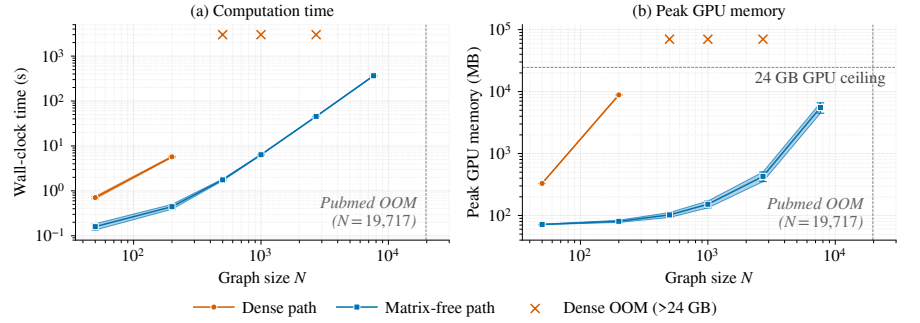


Fig. 5: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond $N=200$; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo ($N=7,650$) at 365 s/5.5 GB.

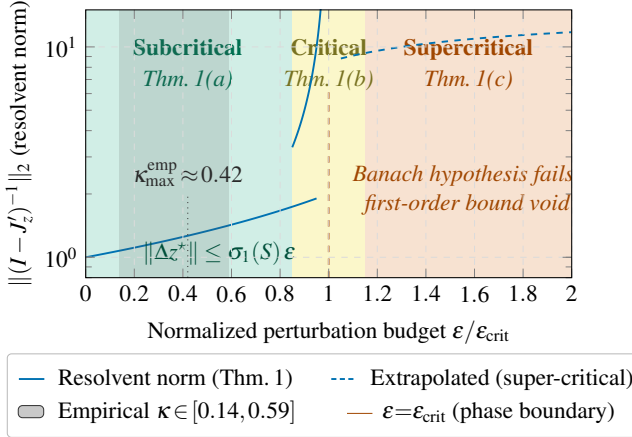


Fig. 6: Three-regime characterisation of theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\epsilon \rightarrow \epsilon_{\text{crit}}$.

S_c from S_K identically (theorem 8). Table V reports the per-architecture characterisation on Cora: IGNN carries theorem 1; the explicit models receive the computational tool without the regime characterisation.

GAT[†] scope. Standard GAT uses A as a binary mask, so $\partial Z/\partial A_{ij}=0$ on existing edges and S_c is undefined; our GAT[†]

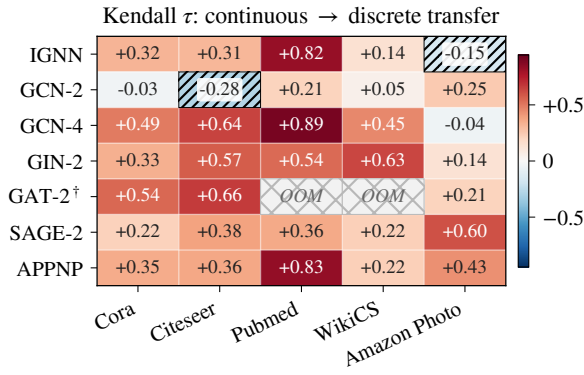
TABLE V: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ : Kendall vs. brute-force single-edge-removal truth. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	$1.01 \pm .00$	$7.6 \pm 0.5 \times$	$+ .32 \pm .04$	$77.5 \pm 1.7 \%$
GCN-2	$1.00 \pm .00$	$3.6 \pm 0.6 \times$	$-.04 \pm .03$	$78.9 \pm 0.9 \%$
GCN-4	$1.02 \pm .00$	$3.4 \pm 0.4 \times$	$+.49 \pm .02$	$78.3 \pm 1.8 \%$
GIN-2	$1.01 \pm .00$	$2.6 \pm 0.1 \times$	$+.33 \pm .03$	$76.6 \pm 1.9 \%$
GAT [†]	$1.01 \pm .01$	$2.1 \pm 0.1 \times$	$+.54 \pm .06$	$77.8 \pm 1.2 \%$
SAGE-2	$1.00 \pm .00$	$1.9 \pm 0.2 \times$	$+.22 \pm .10$	$77.5 \pm 1.5 \%$
APPNP	$1.02 \pm .00$	$3.4 \pm 0.3 \times$	$+.35 \pm .01$	$82.2 \pm 0.6 \%$

modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij} \alpha_{ij} W h_j$), restoring differentiability and yielding tightness 0.99, AtkAdv $4.4 \times$, $\tau = +0.36$. Binary-mask architectures (hard attention, max/min aggregation, GATv2 [47]-style dynamic-attention masking) fall outside the framework.

Robust backbones. Under the same $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite [13] attains $\tau = +0.37/+0.54$ (Cora/Citeseer) and GNNGuard-lite [14] $+0.10/+0.53$, matching or exceeding IGNN; the cap is a precondition, since without it $\kappa > 1$ voids theorem 1.

Cross-dataset transfer. Figure 7 reports τ over 5 datasets and 7 architectures (330 runs); 29/33 cells positive (one-sided sign test $p < 10^{-5}$). Cold cells (GCN-2 on sparse citation



[†] GAT-2: 2-layer attention; OOM on Pubmed and WikiCS.

Fig. 7: Continuous-to-discrete transfer τ across 7 architectures and 5 datasets (10 seeds, 330 runs). Cell values are mean Kendall τ ; buckets: strong ($\tau > 0.5$), moderate (0.3–0.5), weak (0–0.3), anti-correlated ($\tau < 0$, diagonal hatch); cross-hatch: GPU OOM. Deeper models (GCN-4, APPNP) and GAT[†] dominate.

graphs, IGNN on dense product graphs at 50 nodes) recover decisively at full-graph scale: on Amazon Photo ($N=7,650$, stratified top- v_{ij} ground-truth sampling) the edge-weighted ranking $A_{ij}v_{ij}$ achieves $\tau = +0.998 \pm 0.001$ against brute-force N-1 damage — a near-perfect rank match that holds across seeds. On heterophilic WebKB [48] GCN-4 attains $\tau = +0.21 \pm 0.44$.

Theorem 7’s sufficient condition holds for 47–62% of edge pairs across architectures; GCN-2’s 2-hop aggregates produce near-uniform first-order shifts and fall outside the sufficient regime, which the framework correctly flags via a low dataset-level τ (fig. 7).

VI. CASE STUDY: POWER FLOW CONTINGENCY

Beyond classification, AEGIS extends to AC power flow on the IEEE benchmark suite: applying S_c to a ContractiveGCN-PF surrogate recovers brute-force N-1 contingency rankings on case14–118 with $\tau = +0.37$ to $+0.62$ and $P@10 = 0.66$ – 0.81 . This is a label-free, one-query result: v_{ij} is derived purely from the GNN’s learned IFT-resolvent $(I - J_z)^{-1}$, so the correspondence with PTDF/LODF [39] (functions of the DC susceptance matrix B) emerges from the learned physics rather than any DC-model knowledge. Recent GNN-PF work targets fidelity [49], [50], [51], [3] and contingency screening [52]; AEGIS is the label-free vulnerability-attribution layer that operates on any such surrogate without contingency labels or model retraining.

Setup. ContractiveGCN-PF is an IGNN-class operator $F(Z, A) = \phi(\hat{A}ZW^T + X_{\text{proj}})$ with spectral-norm-constrained W , trained on five IEEE test cases [37] with a 2-head readout $(\Delta|V|, \Delta\theta)$. Training data are 2,000 load samples per case (70–130% of nominal, *uniform load scaling only*, a narrow envelope that does not cover seasonal peaks, dispatch shifts, or renewable ramps) from PandaPower’s [38] AC Newton-Raphson solver; inputs are binary adjacency and 5 bus features (slack/PV indicators, P , Q , $|V|$). Ground truth: ℓ_2 voltage-angle deviation per AC contingency. For case14–118 we use full-graph dense analysis; for case300 a 200-node BFS subgraph.

TABLE VI: AEGIS on IEEE power-flow benchmarks (10 seeds). Model quality: per-unit RMSE for $|V|$ and angle θ (std < 0.003). τ : Kendall correlation against brute-force N-1; P@10: top-10 precision. Case14–118 is the headline operating range; case300 is included as a model-capacity stress-test where the GNN surrogate has not yet converged to operational physics.

Case	N	$ E $	Model (p.u.)		AEGIS	
			$ V $	θ	τ	P@10
<i>Credible operating range</i>						
case14	14	20	.007	.020	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	$+.62_{\pm.11}$	$.81_{\pm.10}$
<i>Scalability stress-test (model un-converged)</i>						
case300 [‡]	300	411	.031	.394*	$+.72_{\pm.03}$	$.87_{\pm.06}$

[‡]200-node BFS subgraph; 1,000 training samples.

* θ RMSE $\approx 22.6^\circ$; not a valid operational data point.

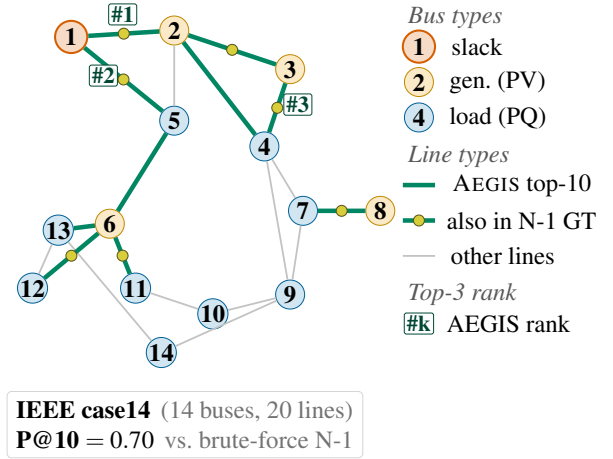


Fig. 8: AEGIS vulnerability ranking on the IEEE 14-bus benchmark (10-seed mean; P@10 = 0.70 for this run, consistent with the 0.74 ± 0.12 in table VI). Bus markers encode physical bus type (slack / generator / load); green edges are the top-10 by v_{ij} with rank tags (#1–#3); yellow midpoint dots mark edges that also appear in the brute-force N-1 top-10 (7 of 10). Top-ranked edges concentrate on the high-flow inter-area tie lines (1–2, 1–5, 7–8), recovering the single-contingency severity pattern from one matrix-free SVD pass.

LODF’s native metric is DC line-flow change; we benchmark against AC voltage-angle truth where rank-order comparison remains informative within LODF’s scope.

table VI reports the envelope ratio and Kendall τ between S_c rankings and brute-force N-1; fig. 8 visualises case14. The envelope is tight in the PF domain (≈ 1.00), and on the credible range (case14–118) ranking correlation is $\tau = +0.37$ to $+0.62$ with P@10=0.66–0.81, identifying 7–8 of the top-10 critical lines. Case300 stress-tests the GNN’s learning capacity, not S_c scalability: at θ RMSE 0.394 p.u. ($\approx 22.6^\circ$) the surrogate has not learned AC physics; its apparent $\tau = +0.72$, P@10=0.87 are conditioned on an unconverged model and excluded from headline numbers. The matrix-free S_c pipeline itself scales to $N=7,650$ (fig. 5); improved PF surrogates would extend the demonstration into the operational range.

Baselines and stability. On the voltage-angle ground

truth, LODF (industry DC screening) attains $\tau=0.44\text{--}0.58$ on credible cases, behind AEGIS's $0.62\text{--}0.67$ on case57/118 (Wilcoxon $p<0.01$, overlap on case14/30); LODF retargeted to thermal-overload reaches $P@10=0.60$ on case57. Ejebe-Wollenberg PI [41] ($n=2$): $\tau=+0.335$, $P@10=0.50$ on case57; $+0.101, 0.30$ on case118. PTDF standalone (no outage correction) is anti-correlated on case57 ($\tau=-0.14$) and near-zero on case118 ($+0.06$), confirming outage redistribution is the operative effect. Runtime: LODF <0.13 s; N-1 $0.1\text{--}2$ s; AEGIS $2\text{--}23$ s. v_1 already recovers $40\text{--}64\%$ of N-2 critical sets via top-5 overlap from a one-query N-1-targeted pass, with pairwise rank stability $+0.40\text{--}+0.78$. *Binary adjacency beats admittance-weighted* ($P@10=0.81$ vs. 0.27 on case118; Spearman admittance vs N-1 severity -0.01 to $+0.15$): a single-line trip removes the line regardless of impedance.

VII. RELATED WORK

We position AEGIS against three threads; each supplies at most one of the three diagnostics AEGIS produces from one S_c object via a single matrix-free pipeline.

Structural attacks. Nettack [1], Mettack [4], GR-BCD/PR-BCD [5] and variants [6] optimise surrogate gradients to flip predictions but provide no analytic globally maximally sensitive direction and no per-node radius. Sober-look studies [16], [17] flag evaluation pitfalls; we follow their adaptive-attack discipline in section V-E (paired Wilcoxon against random masking; an adaptive-attacker column in defense ablation).

Certified robustness and defense. Convex relaxations [12], smoothing [7], [8], [9], [53], collective certificates [10], and IBP [11] certify per-node robustness but do not identify vulnerability-driving edges. Empirical defenses [13], [15], [14] harden models without surfacing per-edge sensitivity. r_v is a first-order threshold, not a sound certificate (theorem 6); the methods are complementary under the prescribed decision rule of table IV.

Implicit networks and influence functions. DEQs [18], IGNN [20], monotone equilibria [54], and well-posedness analyses [27] address *input* sensitivity $\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$ via the resolvent $(I-J_z)^{-1}J_A$ and the unrolled Jacobian S_K . The construction descends from the Koh-Liang influence-function lineage [55] and from implicit-differentiation hyperparameter optimisation [26], [22], but those lines apply IFT to feature- or weight-perturbations of a loss; AEGIS applies it to a *structural-parameter* perturbation of the equilibrium state, projects onto the edge subspace via P_c , and obtains the SVD-optimal direction in a single matrix-free pass rather than via iterative reverse-mode. GNN explainers [56] target prediction fidelity (which subgraph rationalises an output), not adversarial vulnerability; classical matrix perturbation theory [30], [25] supplies the pseudospectral toolkit.

VIII. CONCLUSION

AEGIS produces three adversarial diagnostics (per-edge rankings, the SVD-optimal attack direction, per-node first-order sensitivity radii) from one analytical object, the constrained sensitivity matrix S_c , computed matrix-free for any GNN with

continuous edge-weight-modulated message passing, with a regime characterisation (theorem 1) that holds as a sufficient safety boundary for contractive implicit models.

Limitations. (i) Radii r_v are first-order thresholds, not probabilistic certificates (theorem 6); standard GAT and binary-mask architectures fall outside the framework (section V-F). (ii) Full-graph matrix-free analysis is the recommended graph-level path: AtkAdv amplifies ($3.3\times/9.8\times$ on Cora/Citeseer) and the edge-weighted ranking $A_{ij}v_{ij}$ achieves $\tau\approx+0.998$ on Amazon Photo ($N=7,650$; section V-F). (iii) Opting into the formal $\varepsilon_{\text{crit}}$ track accepts a $\sim 6\%$ accuracy cost, on par with certified-robustness trade-offs. (iv) Operator-grade power-flow deployment, extending case14-118's $\tau=+0.37\text{--}+0.62$ to admittance-weighted edges and broader envelopes, is a natural follow-up. (v) Insertion attacks (Nettack-class) are deliberately scoped out; the S_c construction extends once the edge basis is enlarged to $\bar{E} \supseteq E$ (section II).

Disclosure protocol. (i) A 90-day coordinated-notification window before public release, addressed to the maintainers of OGB graph-robustness benchmarks and to the GR-BCD / Mettack / AGNNCert authors. (ii) Attack-direction code (Algorithm 1 steps 8-9) gated behind institutional-affiliation review mediated by our institutional research-ethics office (named in the camera-ready). (iii) The diagnostic-only path (r_v and v_{ij} scores; no SVD reconstruction) released unconditionally, since it cannot directly synthesise a perturbation. Power-grid artefacts use only public IEEE benchmarks [37].

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